Matrix functions and stability

Complete Crouzeix conjecture

Difficulty: extreme **Importance:** broadly interesting

Status: Lean verified

Topic: Matrix-valued polynomial functional calculus

Rating rationale: Historical ratings retained. The uniform bound must survive arbitrary matrix amplification, beyond the scalar functional calculus; it would give sharp bounds for block matrix functions and connect numerical-range estimates to dilation and similarity theory.

Resolution — independently audited 8 October 2026

OpenAI's mathematics catalog, family 325 reports a proof of the sharp constant-two inequality for finite matrix-valued polynomials, and a stronger version for bounded operators on arbitrary complex Hilbert spaces. Two independent informal AI-agent audits read all 12 pages of the direct proof manuscript, checked its central positive-kernel and block-matrix arguments, and found no gap. Theorem 1.1 covers all base and coefficient sizes and point or line-segment numerical ranges. Its tensor-factor order differs from the block notation below only by a unitary flip. This establishes the original MF-23 target under the catalog's informal-audit rule. The stronger Hilbert-space manuscript was not audited here. The problem ID, canonical path and statement are retained.

At immutable OpenAI revision *adc7f1241b42e322a6451854ab7e4b4c146bf78a*, the project publishes a Lean formalization scope record. Its finite-matrix proof source declares *OAI.DirectCrouzeix.complete_crouzeix : UniversalBound 2*. The Comparator challenge defines *UniversalBound* with arbitrary positive matrix sizes, degree and complex coefficients, using the Euclidean operator norm and the supremum over the numerical range. The formalization catalog maps the declaration to this challenge, and its Comparator configuration permits only Lean's standard foundational axioms.

The 8 October evidence review found no dated successful kernel build for the external source at that time. The subsequent local bridge, independent statement review, and isolated Linux verification are in the evidence section. No external human peer review is asserted.

Lean proof and verification evidence

Reviewed 13 October 2026.

At immutable repository revision *bca7904b*, *NLA.MF23.canonical_crouzeix_proved* proves the local canonical *Target*; *Solution.lean* exports the same target as *NLA.MF23.canonical_crouzeix*. The independent statement and bridge reviews compare that target with the original block polynomial inequality: all positive matrix and block sizes, every degree and complex coefficient, the full numerical range, Euclidean operator norms, and the exact constant two are retained. The proof imports pinned upstream *complete_crouzeix* and uses a norm-preserving tensor-factor swap; it adds no mathematical premise.

The source-locked project pins Lean **4.34.1**, Mathlib *d13f23b723b8*, LeanCert *7f91b6eb3567*, and the upstream source revision recorded in the lock. From *matrix-functions-and-stability/MF-23/lean*, run *lake build Challenge Solution*. The 10 October isolated Linux Comparator/LeanCert job and log passed the selected export and kernel trust; its transitive axioms were only *propext*, *Classical.choice*, and *Quot.sound*. The receipt records the verified source closure and reproduction details. This catalog reviewed that public verification record; it did not rerun Lean locally for this status update.

Context and notation

All matrices are complex, and all norms are Euclidean operator norms. For $A \in \mathbb{C}^{n\times n}$ define

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n,\ x^*x = 1\}.$$

If $F(z) = [f_{ij}(z)]_{i,j=1}^m$ has polynomial entries, let $F(A) = [f_{ij}(A)]_{i,j=1}^m$, an $mn \times mn$ block matrix.

Problem statement

Prove or disprove that, for every $n, m \geq 1$, every $A \in \mathbb{C}^{n\times n}$ and every matrix polynomial $F \in \mathbb{C}^{m\times m}[z]$,

$$\|F(A)\|_2 \leq 2 \max_{z\in W(A)} \|F(z)\|_2.$$

This is the complete, rather than scalar, numerical-range spectral-set question. It controls simultaneous polynomial functions and hence block approximation errors with one constant independent of matrix and block sizes.

References

- Michel Crouzeix, *Numerical range and functional calculus in Hilbert space*, *Journal of Functional Analysis* **244**(2), 668–690 (2007). Original source for the complete conjecture, as explicitly attributed in the next source.
- Per Åhag, Rafał Czyż and Jani Virtanen, *Square Functions, Complete Crouzeix Conjecture in Dimension Three, and the Clouâtre–Ostermann–Ransford conjecture*, arXiv:2608.27346v3, **9 September 2026**, Introduction, equation **(1.2)** and **Theorem 1.1**. This current version states the exact target and explicitly distinguishes it from the scalar proofs. It claims the complete bound for $n \leq 3$. Its counterexample reduction leaves only 2×2 matrix polynomials to check when $n = 4$.
- Michel Crouzeix and César Palencia, *The numerical range is a $(1 + \sqrt{2})$-spectral set*, *SIAM Journal on Matrix Analysis and Applications* **38**(2), 649–655 (2017). The complete universal upper bound is $1+\sqrt{2}$.

Earlier scope and status check (11 September 2026)

The target is source-stated, not an editorial extension of the scalar conjecture. The August 2026 scalar proof claims by Jin and by Lorist–Schwenninger do not settle it: the September 9 primary manuscript explicitly addresses this distinction. The $n = 3$ result was then a recent preprint claim; the full displayed target remained unresolved even if that claim was accepted. Searches through 11 September 2026 for complete Crouzeix proofs, matrix-valued Crouzeix inequalities and later versions found no full solution. The version was inspected as both HTML and PDF; the inequality is on PDF page 2, immediately before Theorem 1.1.